IEDA 4000H

Homework 2 Solutions

1. In Portfolio Omtimization, one is interested in optimally allocating a certain amount of money into a set of N assets (stocks), where the portfolio usually denoted via $\boldsymbol{w} \in \mathbb{R}^N$ is the normalized money invested in each asset (such that $\sum_{i=1}^{N} w_i = 1$). In Modern Portfolio Theory known after Harry Markowitz, the portfolio design is based on maximizing the return while minimizing the risk. This scheme, also known as Mean-Variance Portfolio (MVP), can be formulated via the following optimization problem:

$$\begin{aligned} \max_{\boldsymbol{w}\in\mathbb{R}^N} \quad & \boldsymbol{\mu}^\top \boldsymbol{w} \\ \text{s.t.} \quad & \boldsymbol{e}^\top \boldsymbol{w} = 1, \\ & \boldsymbol{w}^\top \Sigma \boldsymbol{w} \le \alpha, \\ & \boldsymbol{w} \ge 0, \end{aligned} \tag{1}$$

where $\boldsymbol{e}$ is the vector of ones, $\boldsymbol{\mu}$ is the expected (mean) returns of the stocks, and Σ is the covariance matrix of the stock returns. The term $\boldsymbol{w}^\top \Sigma \boldsymbol{w}$ represents the portfolio risk (variance), the square root of which is called volatility. Hence, the objective is to maximize the profit while minimizing the risk, with the parameter α controlling the accepted level of risk. For each solution $\boldsymbol{w}_\alpha^*$, we define the following quantities:

- Portfolio Expected Return: $\mathrm{PER}(\alpha) = \boldsymbol{\mu}^\top \boldsymbol{w}_\alpha^*$,
- Portfolio Volatility: $\mathrm{PV}(\alpha) = \sqrt{\boldsymbol{w}_\alpha^{*\top} \Sigma \boldsymbol{w}_\alpha^*}$,
- Sharpe Ratio: $\mathrm{SR}(\alpha) = \frac{\mathrm{PER}(\alpha)}{\mathrm{PV}(\alpha)}$.

(a) Argue that whether (1) is a convex optimization problem or not.

(b) Let

$$\boldsymbol{\mu} = \begin{pmatrix} 0.06 \\ 0.01 \\ 0.03 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 1.0 & 0.02 & -0.01 \\ 0.02 & 1.0 & -0.1 \\ -0.01 & -0.1 & 1.0 \end{pmatrix}.$$

Use CVXPY to solve (1) for $\alpha \in \{4, 6, \ldots, 14\} \times 10^{-1}$. Plot the efficient frontier, i.e. the diagram of $\mathrm{PER}(\alpha)$ versus $\mathrm{PV}(\alpha)$.

(c) For which value of α in (b), is the Sharpe Ratio maximized?

Solution:

(a) The objective function $\boldsymbol{\mu}^\top \boldsymbol{w}$ is linear (hence convex), the equality constraint $\boldsymbol{e}^\top \boldsymbol{w} = 1$ is affine, the inequality constraint $\boldsymbol{w}^\top \Sigma \boldsymbol{w} \le \alpha$ is convex since Σ is a covariance matrix (hence positive semidefinite), and the inequality constraint $\boldsymbol{w} \ge 0$ is convex. Therefore, (1) is a convex optimization problem.

(b) Here is the Python code using the CVXPY library to solve (1):

```python
import cvxpy as cp
import numpy as np
import matplotlib.pyplot as plt
from IPython.display import display, Markdown

# Data
mu = np.array([0.06, 0.01, 0.03])
Sigma = np.array([[1.0, 0.02, -0.01], [0.02, 1.0, -0.1],
    [-0.01, -0.1, 1.0]])
e = np.ones(3)
alphas = np.arange(0.4, 1.5, 0.2)

# Lists to store results
per = []
pv = []

# Solve for each alpha
for alpha in alphas:
    w = cp.Variable(3)
    objective = cp.Maximize(mu @ w)
    constraints = [e @ w == 1, w >= 0, cp.quad_form(w, Sigma)
        <= alpha]
    problem = cp.Problem(objective, constraints)
    problem.solve(solver=cp.ECOS)

    if problem.status == cp.OPTIMAL:
        w_star = w.value
        per.append(mu @ w_star)
        pv.append(np.sqrt(w_star @ Sigma @ w_star))
        print(f"For alpha = {alpha:.1f}, the optimal w = {
            w_star}.")
    else:
        print(f"Optimization failed for alpha = {alpha}")

# Plot PER vs PV
plt.plot(pv, per, '-o')
plt.xlabel('Portfolio Volatility (PV)')
plt.ylabel('Portfolio Expected Return (PER)')
plt.title('Efficient Frontier')
```

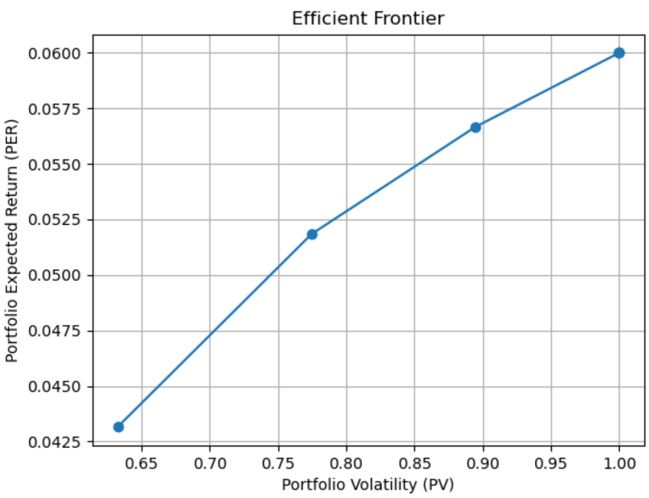
```python
plt.grid(True)
plt.show()

# Compute Sharpe Ratio
sharpe_ratios = [per[i] / pv[i] for i in range(len(alphas))]
max_sharpe_idx = np.argmax(sharpe_ratios)
max_sharpe_alpha = alphas[max_sharpe_idx]
max_sharpe_value = sharpe_ratios[max_sharpe_idx]

print(f"Maximum Sharpe Ratio: {max_sharpe_value:.4f} at alpha
    = {max_sharpe_alpha:.1f}")
```

- For $\alpha = 0.4$, the optimal $w = [0.537957270.149317960.31272477]$.
- For $\alpha = 0.6$, the optimal $w = [7.27991658e-017.27975439e-092.72008335e-01]$.
- For $\alpha = 0.8$, the optimal $w = [8.88574449e-018.72372185e-081.11425464e-01]$.
- For $\alpha = 1.0$, the optimal $w = [9.99999889e-011.62754468e-089.47500719e-08]$.
- For $\alpha = 1.2$, the optimal $w = [9.99999998e-014.27567903e-101.59311803e-09]$.
- For $\alpha = 1.4$, the optimal $w = [9.99999999e-012.67151268e-101.00473958e-09]$.

The efficient frontier is plotted as follows:



(c) The Sharpe Ratio is maximized at $\alpha = 0.4$.

2. Let $\mathcal{X} \subseteq \mathbb{R}^n$ be a closed convex set, θ and f be convex functions on $\mathbb{R}^n$. Suppose that θ is not necessarily smooth. Let f be differentiable on some open set that contains $\mathcal{X}$. Assume that the solution set of the minimization problem $\min\{\theta(\boldsymbol{x}) + f(\boldsymbol{x}) \mid \boldsymbol{x} \in \mathcal{X}\}$ is nonempty. Show that

$$\boldsymbol{x}^* \in \arg\min\{\theta(\boldsymbol{x}) + f(\boldsymbol{x}) \mid \boldsymbol{x} \in \mathcal{X}\} \tag{2a}$$

if and only if

$$\boldsymbol{x}^* \in \mathcal{X}, \quad \theta(\boldsymbol{x}) - \theta(\boldsymbol{x}^*) + (\boldsymbol{x} - \boldsymbol{x}^*)^\top \nabla f(\boldsymbol{x}^*) \geq 0, \quad \forall \boldsymbol{x} \in \mathcal{X}. \tag{2b}$$

Hint: Suppose that (2a) holds. Define $\boldsymbol{x}_\alpha = (1-\alpha)\boldsymbol{x}^* + \alpha\boldsymbol{x}$ for any $\alpha \in (0, 1]$. Then, for any $\boldsymbol{x} \in \mathcal{X}$, we have the following inequality:

$$\frac{\theta(\boldsymbol{x}_\alpha) - \theta(\boldsymbol{x}^*)}{\alpha} + \frac{f(\boldsymbol{x}_\alpha) - f(\boldsymbol{x}^*)}{\alpha} \geq 0.$$

Solution: Suppose that (2a) holds. Define $\boldsymbol{x}_\alpha = (1-\alpha)\boldsymbol{x}^* + \alpha\boldsymbol{x}$ for any $\alpha \in (0, 1]$. Then, for any $\boldsymbol{x} \in \mathcal{X}$, we have the following inequality:

$$\frac{\theta(\boldsymbol{x}_\alpha) - \theta(\boldsymbol{x}^*)}{\alpha} + \frac{f(\boldsymbol{x}_\alpha) - f(\boldsymbol{x}^*)}{\alpha} \geq 0.$$

By the convexity of θ, we have

$$\theta(\boldsymbol{x}_\alpha) \leq (1-\alpha)\theta(\boldsymbol{x}^*) + \alpha\theta(\boldsymbol{x}),$$

which implies

$$\theta(\boldsymbol{x}) - \theta(\boldsymbol{x}^*) \geq \frac{\theta(\boldsymbol{x}_\alpha) - \theta(\boldsymbol{x}^*)}{\alpha}, \quad \forall \alpha \in (0, 1].$$

Substituting this into the first inequality, we obtain

$$\theta(\boldsymbol{x}) - \theta(\boldsymbol{x}^*) + \frac{f(\boldsymbol{x}_\alpha) - f(\boldsymbol{x}^*)}{\alpha} \geq 0, \quad \forall \alpha \in (0, 1].$$

Since $f(\boldsymbol{x}_\alpha) = f(\boldsymbol{x}^* + \alpha(\boldsymbol{x} - \boldsymbol{x}^*))$, taking the limit as $\alpha \to 0_+$ yields

$$\theta(\boldsymbol{x}) - \theta(\boldsymbol{x}^*) + \nabla f(\boldsymbol{x}^*)^\top (\boldsymbol{x} - \boldsymbol{x}^*) \geq 0, \quad \forall \boldsymbol{x} \in \mathcal{X},$$

which is equivalent to (2b). For the converse, see Tutorial 3, slide 19.

3. Consider the following optimization problem:

$$\begin{aligned} \min_{\boldsymbol{x} \in \mathbb{R}^n} \quad & \boldsymbol{x}^\top \mathrm{W} \boldsymbol{x} \\ \text{s.t.} \quad & x_i^2 = 1, \quad i = 1, \ldots, n, \end{aligned} \tag{3}$$

where $\mathrm{W} \in \mathbb{R}^{n \times n}$ is a symmetric matrix. Let $\boldsymbol{e}$ denote the vector of all ones and for $\boldsymbol{\lambda} \in \mathbb{R}^n$,

$$\operatorname{diag}(\boldsymbol{\lambda}) = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{pmatrix}.$$

(a) Write the Lagrangian function of (3).

(b) What is the Lagrangian dual of (3)?

(c) Let $\mathrm{W} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$. Solve (3) and its Lagrangian dual problem. Find the duality gap.

Solution:

(a) The Lagrangian function of (3) is

$$L(\boldsymbol{x}, \boldsymbol{\lambda}) = \boldsymbol{x}^\top \mathrm{W} \boldsymbol{x} - \sum_{i=1}^n \lambda_i (x_i^2 - 1) = \boldsymbol{x}^\top \big(\mathrm{W} - \operatorname{diag}(\boldsymbol{\lambda})\big) \boldsymbol{x} + \boldsymbol{e}^\top \boldsymbol{\lambda}.$$

(b) The Lagrangian dual function is

$$h(\boldsymbol{\lambda}) = \inf_{\boldsymbol{x} \in \mathbb{R}^n} L(\boldsymbol{x}, \boldsymbol{\lambda}) = \inf_{\boldsymbol{x} \in \mathbb{R}^n} \Big(\boldsymbol{x}^\top \big(\mathrm{W} - \operatorname{diag}(\boldsymbol{\lambda})\big) \boldsymbol{x} + \boldsymbol{e}^\top \boldsymbol{\lambda} \Big).$$

If $\mathrm{W} - \operatorname{diag}(\boldsymbol{\lambda}) \succeq 0$, then $h(\boldsymbol{\lambda}) = \boldsymbol{e}^\top \boldsymbol{\lambda}$; otherwise, $h(\boldsymbol{\lambda}) = -\infty$. Therefore, the Lagrangian dual problem is

$$\begin{aligned} \max_{\boldsymbol{\lambda} \in \mathbb{R}^n} \quad & \boldsymbol{e}^\top \boldsymbol{\lambda} \\ \text{s.t.} \quad & \mathrm{W} - \operatorname{diag}(\boldsymbol{\lambda}) \succeq 0. \end{aligned}$$

(c) For $\mathrm{W} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, the problem (3) can be written as

$$\begin{aligned} \min_{x_1, x_2 \in \mathbb{R}} \quad & 2 x_1 x_2 \\ \text{s.t.} \quad & x_1^2 = 1, \quad x_2^2 = 1. \end{aligned}$$

The feasible set is $\{(-1,-1),(-1,1),(1,-1),(1,1)\}$, and the optimal value is -2 attained at $(-1,1)$ and $(1,-1)$. The Lagrangian dual problem is

$$\begin{aligned} \max_{\lambda_1,\lambda_2\in\mathbb{R}} \quad & \lambda_1+\lambda_2 \\ \text{s.t.} \quad & \begin{pmatrix} -\lambda_1 & 1 \\ 1 & -\lambda_2 \end{pmatrix} \succeq 0. \end{aligned}$$

The constraint is equivalent to $\lambda_1 \le 0$ and $\lambda_1\lambda_2 \ge 1$. Therefore, the optimal value of the dual problem is -2 attained at $(\lambda_1,\lambda_2)=(-1,-1)$. Since both primal and dual optimal values are -2, the duality gap is 0.